BRD(BUSINESS REQUIRED DOCUMENTS)

Project Description

The Indian Premier League (IPL) generates a vast amount of match and ball-by-ball data every season. While the data contains valuable information regarding teams, players, venues, batting, bowling, and match outcomes, it is difficult for cricket enthusiasts and analysts to extract meaningful insights directly from raw datasets.

The objective of this project is to develop an interactive IPL Analytics Dashboard using SQL and Microsoft Power BI. SQL will be used to clean, transform, and prepare the data, while Power BI will be used to create interactive dashboards that display key performance indicators (KPIs), team statistics, player performance, and season-wise insights. The dashboard will help users analyze historical IPL performance through easy-to-understand visualizations.

Project Name

IPL Data Analysis & Interactive Dashboard

Client

Rachna

Start Date

27 June 2026

End Date

(To be filled)

Prepared By

Krishna Gopal

Problem Statement

- Which team won the IPL tournament in each season?
- Which team finished as the runner-up?
- How many sixes and fours have been hit across all IPL seasons?

- Which player won the Orange Cap in each season?
- Which player won the Purple Cap in each season?
- Which teams participated in every IPL season?
- Which venues hosted the highest number of IPL matches?
- How many centuries and half-centuries were scored?
- Which team has won the most matches?
- What are each team's total wins, losses, ties, and points?
- How has team performance changed across different IPL seasons?

Objectives

Tournament Analysis

- Identify IPL Winner Team for each season.
- Identify IPL Runner-up Team for each season.
- Calculate total matches played across all seasons.
- Calculate total participating teams.

Batting Analysis

- Calculate Total Sixes.
- Calculate Total Fours.
- Calculate Total Centuries.
- Calculate Total Half-Centuries.
- Identify Orange Cap Winner.

Bowling Analysis

- Identify Purple Cap Winner.
- Calculate leading wicket-takers.

Team Analysis

- Calculate Matches Played.
- Calculate Matches Won.
- Calculate Matches Lost.
- Calculate Tied Matches.
- Calculate Total Points.
- Display Team Logo and Team Information.

Venue Analysis

- Calculate Total Venues Used.
- Identify venues hosting the maximum number of matches.

Dashboard Development

- Build an interactive Power BI dashboard.
- Enable filtering by Season, Team, Venue, and Player.
- Present KPIs using Cards, Charts, Tables, and Slicers.

Scope

Dashboard KPIs

Primary KPIs

- Winner Team
- Runner-up Team

Secondary KPIs

- Total Matches
- Teams Participated
- Total Sixes
- Total Fours
- Total Centuries
- Total Half-Centuries
- Total Venues

Player KPIs

- Orange Cap Winner
- Purple Cap Winner
- Player Name
- Player Image

Team KPIs

- Team Logo
- Team Name
- Matches Played
- Won

- Lost
- Tie
- Total Points

Dashboard Features

- Season Filter
- Team Filter
- Venue Filter
- Interactive Charts
- Drill-through Analysis

Stakeholders

- Cricket Fans
- Sports Analysts
- Business Analyst
- Data Analyst
- Dashboard Users

Dataset

- matches.csv
- deliveries.csv

Tech Stack

- PostgreSQL
- Microsoft Power BI
- Microsoft Excel (for validation)

Deliverables

- Clean SQL Database
- SQL Queries for KPI Calculation
- Power BI Dashboard
- Business Requirement Document

- Project Documentation

Success Criteria

- Winner and Runner-up calculated correctly.
- Orange Cap and Purple Cap winners displayed accurately.
- Total Sixes, Fours, Centuries, Half-Centuries, Teams, Venues, and Matches calculated correctly.
- Team statistics (Won, Lost, Tie, Points) displayed correctly.
- Dashboard filters work correctly.
- Dashboard loads efficiently and provides interactive analysis.

Approval Status

Client Signature: _____

Project Guide: _____